9th International XR-Metaverse Conference 2024 (XRM 2024)
Busan, South Korea, May 20-22, 2024
Copyright©2024 IAITI

Design and Implementation of a Digital Twin System for Real-time Monitoring and Management of Tree Health in the Tree Doctor EMR

Eu-Tteum Baek and Hye-Seong Yoon

AI&Bigdata, Honam University, South Korea
geodo100@gmail.com, remisen326@naver.com

Abstract

Our study presents the fusion of Digital Twin technology into EMR for effective tree management. By integrating real-time monitoring, visualization tools, and predictive analytics, tree managers gain empowered decision-making abilities. This approach enhances data synchronization, offering comprehensive insights for proactive tree health management. Our system transforms arboriculture prac-tices, promising a sustainable future for tree preservation and health.

Keywords: digital twin system; EMR; Tree doctor

1 Introduction

While EMR systems aim to centralize tree-related information, integrating them seamlessly within arboriculture presents challenges. The complex nature of tree condition data, coupled with limitations in existing EMR frameworks, makes it tough to visualization, analysis, and prediction of crucial parameters. This integration struggle results in fragmented data, inadequate visualization tools, and a lack of predictive analytics, hindering accurate diagnosis and proactive management of tree health [1].

This study proposes a solution by merging EMR systems designed for arborists with Digital Twin technology. Digital Twins replicate physical entities in a virtual space, offering a precise representation [2-3]. Leveraging advancements like IoT and sensors, these digital replicas bridge the gap between the physical and virtual realms, enabling real-time tracking and monitoring of tree conditions. Incorporating Digital Twins into EMRs holds the potential to address these challenges, providing enhanced insights and tools for managing tree health.

this paper introduces a novel concept: the integration of Digital Twins within EMRs to alter tree care practices. It outlines a strategy for a customized Digital Twin tailored for EMRs, addressing the complexities inherent in current systems. The focus is on smart integration, enabling improved visualization of tree data and showcasing how this innovation can redefine tree monitoring, disease diagnosis, and predictive capabilities. Through rigorous validation, this research aims to demonstrate the viability of this approach, trying for a more robust system to manage tree health.

2 Methodology

The paper focuses on developing and integrating a specialized digital twin system in the EMR setup made for tree managers. It aims to make it easier to handle, see, and predict tree health data for better decision-making. The main goal is to come up with ways to connect existing EMR systems with the Digital Twin setup. These ways ensure smooth sharing of information, using standard methods to make it simple for different parts of the system to work together.

The subsequent phase emphasizes implementing the Digital Twin framework, integrating intuitive visualization tools to represent tree health metrics and disease patterns comprehensively. Predictive analytics, utilizing machine learning or statistical modeling, anticipates disease outbreaks and future tree conditions from historical and real-time data.

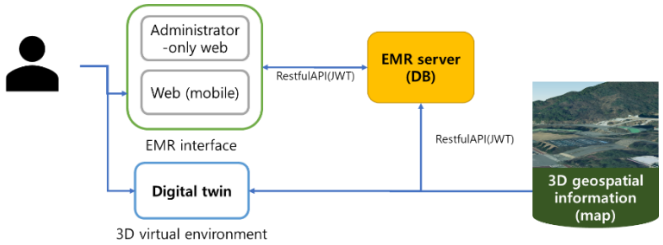

Fig. 1. Schematic diagram of EMR digital twin.

2.1 Whole process of EMR digital twin

Figure 1 illustrates the configuration diagram outlining the proposed system. The EMR system interface empowers tree managers to update and modify tree information within the EMR system. This interface grants users the capability to add new trees, or edit and remove existing tree details seamlessly. The Tree Doctor System's server hosts a dedicated database (DB) housing comprehensive tree-related information. Any updates made through the EMR interface promptly synchronize with and enrich the server's DB. Concurrently, the server reflects the updated data within its DB and triggers the transmission of fresh information to the digital twin upon request through an API. Within the digital twin, the server's DB information is accessed through an API. The twin employs a specific endpoint to solicit information from the server, assimilating the retrieved data into its simulated environment. This integration allows for real-time data synchronization between the server's DB and the digital twin, ensuring consistent and updated information exchange.

3 Results and Discussion

A prototype of a tree doctor digital twin system was developed using the Unity3D engine. Within the simulated environment, tree managers possess the capability to enhance their tree information management by efficiently updating and visualizing tree data. This feature enables effective monitoring and assessment of crucial tree-related information. Figures 2 and 3 serve as concrete illustrations of this technology's applications. These depictions encompass crucial elements such as tree selection, the impact assessment of tree diseases, interrelations between trees, and detailed tree descriptions. This visual representation serves as a valuable resource aiding professional in acquiring essential insights necessary for effective decision-making in their practical work. This anticipatory preparation significantly enhances operational efficiency, which, in turn, is expected to positively influence the overall health and preservation of trees. The amalgamation of digital twin technology and simulation stands as a testament to the potential enhancement of tree management methodologies, promising better outcomes and sustainability in arboricultural practices.

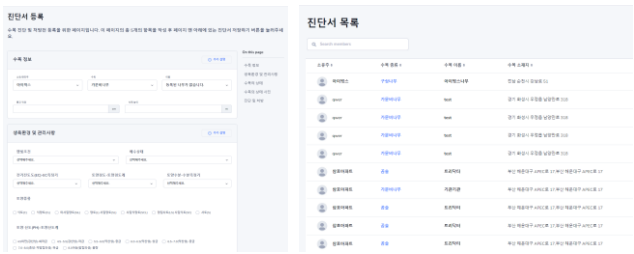

Fig. 2. Illustration Depicting EMR Service Implementation for Arborists.

(a) (b) (c) (d)

Fig. 3. Exemplars of 3D Virtual Digital Twins for Arborists. (a) tree selection, (b) the Impact of Tree Plague, (c) t ree relationship and (d) tree description

4 Conclusion

Our study presents the integration of Digital Twin technology into EMR systems marks a significant step forward in dealing with the complexities of tree health management. This study's approach, featuring real-time monitoring, visualization tools, and predictive analytics, empowers tree managers to make informed decisions. The proposed method is expected to enhance operational efficiency in tree management and contribute to the sustainability of tree conservation.

References

1. Croatti, A., Gabellini, M., Montagna, S., & Ricci, A. (2020). On the integration of agents and digital twins in healthcare. Journal of Medical Systems, 44, 1-8.
2. Turab, M., & Jamil, S. (2023). A Comprehensive Survey of Digital Twins in Healthcare in the Era of Metaverse. BioMedInformatics, 3(3), 563-584.
3. Hassanien, A. E., Darwish, A., & Snasel, V. (Eds.). (2022). Digital Twins for Digital Transformation: Innovation in Industry. Springer International Publishing.
4. Helbing, Dirk, and Javier Argota Sánchez-Vaquerizo. "Digital twins: potentials, ethical issues and limitations." Handbook on the politics and governance of Big Data and Artificial Intelligence. Edward Elgar Publishing, 2023. 64-104.
5. Jiang, Xuetao, et al. "Forestry digital twin with machine learning in Landsat 7 data." Frontiers in Plant Science 13 (2022): 916900.
6. Jawhari, B., Ludwick, D., Keenan, L., Zakus, D., & Hayward, R. (2016). Benefits and challenges of EMR implementations in low resource settings: a state-of-the-art review. BMC medical informatics and decision making, 16(1), 116

Acknowledgements

This work is resulted from the AI Convergence University Support Project in Honam University financially supported by the Ministry of Science and ICT.